Experiment 3: Network Switching Subsystem Flowchart

Aim

To develop a flowchart in Figma depicting the operation of the Network Switching Subsystem (NSS) in a mobile cellular system, showing the process of call initiation, authentication, switching, routing, and call termination.

Algorithm

Open Figma and create a new design file.

Create a flowchart frame with the title “Network Switching Subsystem in Mobile Cellular System.”

Start the flowchart with the Mobile Station (MS).

Show the Base Station Subsystem (BSS) receiving the call request from the mobile station.

Send the request to the Mobile Switching Centre (MSC).

The MSC checks the subscriber information using the Home Location Register (HLR).

Use the Authentication Centre (AuC) to authenticate the subscriber.

Use the Visitor Location Register (VLR) to obtain temporary subscriber/location information.

If authentication is successful, the MSC establishes and routes the call.

If the destination is another network, route the call through the appropriate Gateway MSC (GMSC).

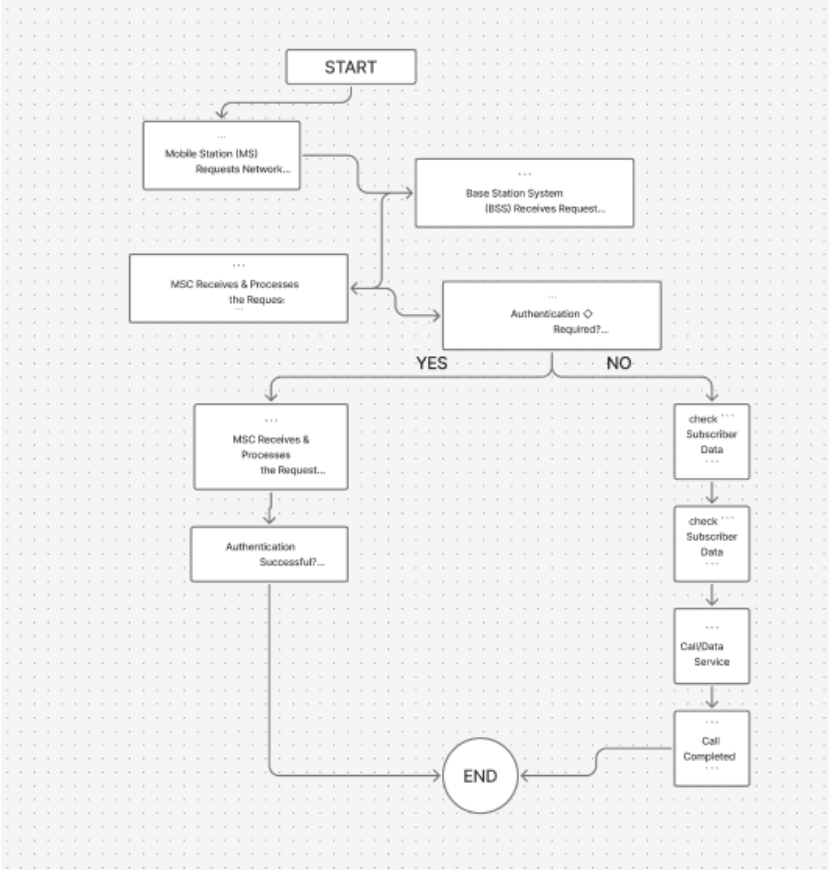
Establish the connection between the calling and receiving users.

During the call, the MSC manages switching and mobility.

When the call ends, release the allocated network resources.

Add arrows and decision symbols in Figma to clearly show the flow.

Use Prototype mode to make important flowchart elements interactive if required.



Result

The Network Switching Subsystem flowchart was successfully developed in Figma. The flowchart represents the complete operation of the NSS, including call initiation, subscriber authentication, HLR/VLR lookup, switching through the MSC, call routing, connection establishment, and resource release. It provides a clear visual representation of how a cellular network handles a mobile call.